

Fernando Sepulveda - CV

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EDUCATION

The University of Texas at El Paso (UTEP)

- Master of Science in Computer Science
- Bachelor of Science in Computer Science

Expected: 05/2027

Graduated: 12/2024

Relevant Coursework

Data Mining | Machine Learning | Deep Learning | Software Eng: Design & Implementation | Matrix Algebra

PROFESSIONAL EXPERIENCE

Mike Loya Center for Innovation and Commerce

El Paso, TX

Graduate Research Fellow

01/2026 – Present

- Mentor student entrepreneurs in the development and commercialization of AI-driven business models, facilitating the transition of graduate-level research into market-ready prototypes
- Evaluate the technical feasibility and scalability of AI startups, providing strategic guidance on software architecture and algorithmic efficiency to optimize product-market fit
- Collaborate with the Institute for Applied AI Innovation (AII) to design and implement outreach strategies, fostering a regional innovation ecosystem for emerging technology ventures

Smart Autonomous Mobility Systems (SAMS) Lab - UTEP

El Paso, TX

Undergraduate Research Student

08/2023 – 12/2024

- Published research on Adaptive Cruise Control for the next generation of autonomous driving to increase fuel efficiency and positive traffic flow with AI and cyber-considerate approaches
- Developed a python script to simulate a 10-car platoon with adaptive cruise control formulas enabling the visualization of driving dynamics demonstrating positive string stability
- Utilized CPLEX and MATLAB to solve optimization problems in car simulations, identifying optimal design parameters and driving strategies to enhance vehicle performance, fuel efficiency, and safety

Kansas City National Security Campus - Honeywell

Kansas City, KS

Student Engineering Intern III – Data Scientist – Q level clearance

06/2024 – 08/2024

- Implemented automated factory operations in the Smart Factory through POLCA systematics; utilizing useful tools such as Dataiku, Hadoop, API integration, SQL, Angular, and Python to gather data and redistribute floor operation efforts
- Implemented Machine Learning knowledge on Large Language Model (LLM) project to generate a security-first search system for gathering documentation and retrieval of sources with an agile-based team
- Effectively contributed towards black belt automation project with impact of projected cost savings upwards of \$250,000 per fiscal year through time efficiency

Oracle Cloud Infrastructure (OCI) - Oracle

Austin, TX

Software Engineer Intern

05/2023 – 08/2023

- Automated the management of hypervisors by leading the implementation of a script written in java enabling the user to organize hosts for a specific region depending on needs such as maximum pool size or pool shape
- Orchestrated a reusable façade class integrating with “ODO” API enabling for RESTful calls between front-end user declarations, façade computation, and API calls for successful segmentation and retrieval of data
- Command line interface was documented on confluence ensuring reusability; the code resulted in improved efficiency for the team by saving up to 50% of previous deployment time

W. M. Keck Center for 3d Innovation - UTEP

El Paso, TX

Undergraduate Research Student

08/2022 – 06/2023

- Developed and maintained Database systems implementations within python built webapp using Flask with PyCharm for cloud-based data management system Hyperthought
- Examined and documented the quality of SLM artifacts by conducting porosity and nominal-to-actual analyses using XCT Machine with VG Studio Software resulting in consistent print grade improvements
- Built a connection between Mechanical Engineering and Computer Science teams by identifying weaknesses between the data collection process resulting in a more consistent development process

Missouri University of Science and Technology (MS&T)

Undergraduate Research Intern (REU)

Rolla, MO

05/2022 – 08/2022

- Developed novel additive manufacturing technique, Laser Foil Printing for advanced materials with the goal of furthering the current understanding of the promising metal foil printing techniques
- Implemented and tested active cooling system for Metallic Glass production using LASERDESK with amorphous composition leading towards achieving 85% efficacy from proposed goals
- Documented on prospective automated printing improvements using algorithm for temperature logging

SKILLS

- **Programming Languages:** Python (Advanced), Java (Advanced), Modern C++ (17/20), SQL, JavaScript, PHP, C, Dart, TypeScript, Shell Scripting/ Bash, ROS2
- **Software & Tools:** Linux (Advanced), Dataiku, Hadoop, Flask, Angular, Git, Slack, Microsoft Suite, Google Workspace, Adobe Suite, Docker, Kubernetes
- **Machine Learning & AI:** AI Fundamentals, Transformer-based LLMs, Convolutional Neural Networks (CNNs), TensorFlow, API Integration, PyTorch, scikit-learn, Pandas & NumPy, Elastic search, SBERT
- **Cloud & Databases:** Oracle Cloud Infrastructure (OCI), Amazon Web Services (AWS), Database Systems (MySQL, Neo4j, PostgreSQL, MongoDB)
- **Engineering & Simulation Tools:** Fusion 360, CURA, SOLIDWORKS, AutoCAD, laserDESK, LabVIEW, MATLAB, VG Studio
- **Development Practices:** Agile Development, API Design, Automation, RESTful Services
- **Languages:** Trilingual — English (C2), Spanish (C2), French (C1), Portuguese (A1)

Honors & Activities

- Mike Loya Center for Innovation and Commerce Fellow – January 2026
- TRAX International Endowed Scholar – August 2024
- NMSU Entrepreneurship 10-week Course Completion – November 2023
- Miners Pitch Competition (1st place) – October 2023
- UNCF Oracle Scholar – June 2023
- HITEC Scholar Recipient – 2022 & 23 & 24
- International SFF Symposium Presenter – August 2022
- MSIPP Scholar – Fall 2022
- Chihuahuas All Star Legacy Scholar – 2021
- Coronado HS PTSA Vocational Scholar – 2021
- MOWW Award | Eagle Scout Award – 2019

PROJECTS/ LEADERSHIP

Machine Learning Research Projects (UTEP)

El Paso, TX

Machine Learning Implementation Applications

08/2022 – Present

- **Biometric Security & Facial Recognition (CNN):** Developed and optimized a Convolutional Neural Network (CNN) using TensorFlow/Keras to evaluate biometric security robustness; implemented adversarial testing protocols to measure accuracy degradation and identify vulnerabilities in facial recognition authentication under variable environmental noise
- **Real-Time Autonomous Vehicle Perception (YOLO):** Architected a high-speed object detection system utilizing the YOLO (You Only Look Once) framework to perform real-time identification and tracking of pedestrians, vehicles, and traffic signs, optimizing inference latency and mean Average Precision (mAP) for safety-critical autonomous navigation.
- **Semantic Segmentation for Lane Navigation:** Implemented an advanced computer vision pipeline leveraging Semantic Segmentation and OpenCV to extract road boundaries and calculate lane curvature; utilized deep learning models to perform pixel-wise classification, enabling precise path planning for autonomous vehicles in complex urban environments.

POST 3d Prints

El Paso, TX

Founder and CEO

08/2022 – 12/2025

- Engineered full-stack marketplace connecting AM providers with consumers using microservices architecture

- Validated market fit through analysis of 30+ customer interviews and iterative development methodology
- Implemented data-driven decision framework, winning first place in competitive venture capital pitch competition

Society of Hispanic Professional Engineers (SHPE/ MAES)

El Paso, TX

Team Leader

08/2021 – 12/2024

- Led outreach initiatives that increased chapter membership by 20%, enhancing the representation of Hispanic professionals in STEM fields
- Organized and facilitated professional development workshops that provided members with essential career skills, resulting in a 15% increase in internship placements

Borderhack Hackathon (Google Developer Student Club)

El Paso, TX

Software lead – UI Specialist

Spring 2023

- Implemented GPT-3.5 API into a community support app built on flutter flow which allowed for sorting of supplies; the app was successful in connecting users with required supplies at a regional level (2nd place)
- Led the team towards victory by deliberating workflow on the task with a very strong time constraint

Coding Interview Club (UTEP)

El Paso, TX

Treasurer

08/2021 – 12/2024

- Managed club finances and supported event planning for weekly coding interview prep sessions focused on algorithmic problem solving.
- Practice vital technical coding skills encompassing knowledge, speed, creativity, and execution
- Network with other students to build relationships in my Computer Science classes resulting in positive group efforts during assignments

IEEE Robotics & Automation Society (UTEP)

El Paso, TX

Programmer

08/2022 – 12/ 2024

- Assist with the research and development of programing for control module in VEX robotics avatar
- Develop knowledge from electrical engineering concepts while networking with other students

STEELE Force Hackathon (STEELE Cons.)

El Paso, TX

Software Architect

Spring 2022

- Collaborated with team to produce 11/14 scored shooter game using JavaScript on Typescript framework
- Overcame the obstacle of working with completely new technologies by using an agile work style and prioritizing the most important tasks to have working code

Publications

1. An Impact Evaluation of Strategic Cyberattacks on Autonomous Vehicles: Safety, Mobility, and Energy Consumption

- Edmonton, Canada – September 2024
- Department of Electrical and Computer Engineering, Department of Computer Science, University of Texas at El Paso
- IEEE International Conference on Intelligent Transportation Systems

Presentations

1. An Impact Evaluation of Strategic Cyberattacks on Autonomous Vehicles: Safety, Mobility, and Energy Consumption

- Las Cruces, NM – April 2022 – [\[1\]](#)
- Department of Electrical and Computer Engineering, University of Texas at El Paso

2. Metal Additive Manufacturing for Advanced Materials by Laser Foil Printing

- Austin, TX – July 2022 – [\[2\]](#)
- Department of Mechanical and Aerospace Engineering, Missouri University of Science and Technology

Certifications

Additive Manufacturing for Business

El Paso, TX

LinkedIn

Fall 2022

Machine Learning Foundations: Linear Algebra

El Paso, TX

LinkedIn

Fall 2022

Python for Engineers and Scientists

El Paso, TX

LinkedIn

Fall 2022